Unit 6 Quiz: Probability KEY

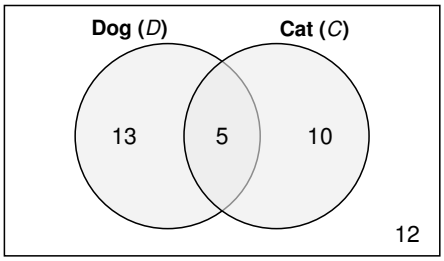
Name _____

Show your work.

6.1 - Venn Diagrams and the Addition Rule

1. **I find a probability and its complement, and I read the symbols $A \cup B$ and $A \cap B$.**

   40 students said if they have a dog (D) and if they have a cat (C).



   (a) Find $P(D)$.

   $$\frac{13+5}{40} = \frac{18}{40} = \frac{9}{20} = 0.45$$

   (b) Find $P(\text{not } D)$.

   $$1 - \frac{9}{20} = \frac{11}{20} = 0.55$$

   (c) What does $D \cap C$ mean in words? Find $P(D \cap C)$.

   Has a dog **and** has a cat. $\frac{5}{40} = \frac{1}{8} = 0.125$

   (d) What does $D \cup C$ mean in words? Find $P(D \cup C)$.

   Has a dog **or** a cat **or** both. $\frac{13+5+10}{40} = \frac{28}{40} = \frac{7}{10} = 0.7$

2. **I use the addition rule, and I decide if two events are mutually exclusive.**

   (a) $P(A) = 0.35$, $P(B) = 0.45$, and $P(A \cap B) = 0.15$. Find $P(A \cup B)$.

   $P(A \cup B) = 0.35 + 0.45 - 0.15 = 0.65$

   Draw one card from a deck of 52. Face cards are J, Q, and K.
   Are the events mutually exclusive? Find the probability.

   (b) a 7 or a queen

   Yes. A card cannot be a 7 and a queen.
   $\frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13} \approx 0.15$

   (c) a club or a face card

   No. The J, Q, and K of clubs are both.
   $\frac{13}{52} + \frac{12}{52} - \frac{3}{52} = \frac{22}{52} = \frac{11}{26} \approx 0.42$

6.2 - The Multiplication Rule

3. **I use the multiplication rule for independent events and the complement for "at least one" questions.**

   A school bus is on time on 90% of days. The days are independent. Look at 3 school days in a row.

   (a) Find P(on time all 3 days).

   $0.9 \cdot 0.9 \cdot 0.9 = 0.729$

   (b) Find P(late all 3 days).

   $0.1 \cdot 0.1 \cdot 0.1 = 0.001$

   (c) Find P(late at least once).

   $1 - P(\text{on time all 3 days}) = 1 - 0.729 = 0.271$

4. **I decide if two events are independent or dependent, and I use the general multiplication rule.**

   A box has 5 red pens and 7 blue pens. You take 2 pens without looking.

   (a) You put the first pen back before you take the second. Find P(red, red).

   Independent. $\frac{5}{12} \cdot \frac{5}{12} = \frac{25}{144} \approx 0.17$

   (b) You keep the first pen out. Find $P(\text{red second} \mid \text{red first})$.

   4 red pens are left out of 11 pens. $\frac{4}{11} \approx 0.36$

   (c) You keep the first pen out. Find P(red, red).

   Dependent. $\frac{5}{12} \cdot \frac{4}{11} = \frac{20}{132} = \frac{5}{33} \approx 0.15$

   (d) Which pair of events is dependent? Circle the letter.

   A Roll a number cube two times.
   B Flip a coin, then spin a spinner.
   C Pick a name from a hat, put it back, and pick again.
   (D) Pick a name from a hat, keep it out, and pick again.

6.3 - Conditional Probability

5. **I find conditional probabilities from a two-way table, and I check if two events are independent.**

80 students said if their earbuds are wired or wireless.

	Wired	Wireless	Total
9th grade	24	16	40
10th grade	12	28	40
Total	36	44	80

(a) Find $P(\text{wireless} \mid \text{10th grade})$.

Only the 40 in 10th grade.
$\frac{28}{40} = \frac{7}{10} = 0.7$

(b) Find $P(\text{10th grade} \mid \text{wireless})$.

Only the 44 with wireless.
$\frac{28}{44} = \frac{7}{11} \approx 0.64$

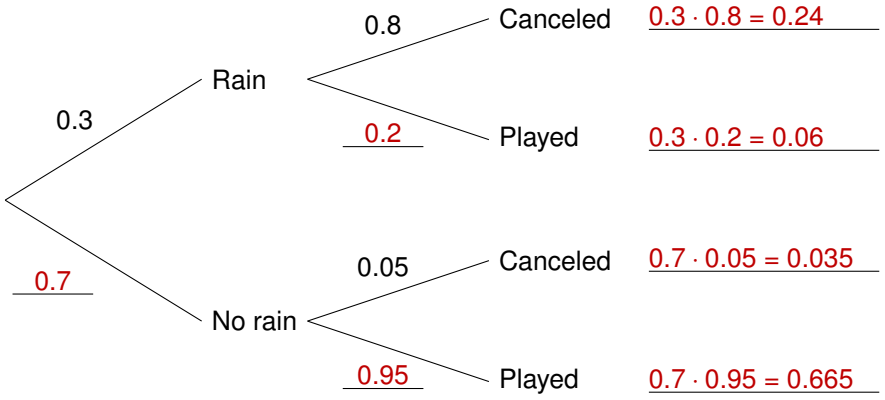
(c) Are having wireless earbuds and being in 10th grade independent? Show your work.

$P(\text{wireless}) = \frac{44}{80} = 0.55$, but $P(\text{wireless} \mid \text{10th grade}) = 0.7$.
They are different, so the events are **not** independent.

6. **I use a tree diagram and the notation $P(A \mid B)$ to find probabilities.**

The chance of rain on Saturday is 30%. If it rains, the chance that the soccer game is canceled is 80%. If it does not rain, the chance that the game is canceled is 5%.

(a) Fill in the tree diagram. Write the missing probabilities and multiply along each path.



(b) Find $P(\text{canceled})$.

Add the two canceled paths. $0.24 + 0.035 = 0.275$

(c) The game was canceled. Find $P(\text{rain} \mid \text{canceled})$.

$\frac{0.24}{0.275} = \frac{48}{55} \approx 0.87$

6.4 - Simulations

7. **I tell theoretical and experimental probability apart, and I explain the law of large numbers.**

   A spinner has 5 equal parts. One part is blue. Lena spins it 50 times. It lands on blue 14 times.

   (a) Find the experimental probability of blue.

   $\frac{14}{50} = \frac{7}{25} = 0.28$

   (b) Find the theoretical probability of blue.

   $\frac{1}{5} = 0.2$

   (c) Lena spins it 5,000 times. What number do you expect the experimental probability of blue to be close to? Why?

   Close to 0.2. The law of large numbers: with more trials, the experimental probability gets closer to the theoretical probability.

8. **I assign random digits and run a simulation to estimate a probability.**

   A soccer team wins 60% of its games. Simulate the next 4 games.
   Digits 0 to 5 = win. Digits 6 to 9 = loss. One trial = 4 digits (4 games).

   (a) Why do the digits 0 to 5 stand for a win?

   0 to 5 is 6 of the 10 digits, and $\frac{6}{10} = 60\%$.

   (b) Use these random digits to run 10 trials. Count the wins in each trial.

   76869 45827 19553 09040 57143 14173 96580 44469

Trial	1	2	3	4	5	6	7	8	9	10
Digits	7686	9458	2719	5530	9040	5714	3141	7396	5804	4469
Wins	0	2	2	4	3	3	4	1	3	2

   (c) Use your simulation to estimate P(at least 3 wins).

   Trials 4, 5, 6, 7, and 9 have 3 or more wins. $\frac{5}{10} = 0.5$

   (d) A different team wins 35% of its games. How would you assign random digits to one game?

   Use pairs of digits. 00 to 34 = win (35 of the 100 pairs). 35 to 99 = loss.

6.5 - Permutations

9. **I use the counting principle and factorials.**

   (a) A bike comes in 4 colors, 3 sizes, and 2 seat styles. How many different bikes are there?

   $4 \cdot 3 \cdot 2 = 24$ bikes

   (b) Find $\frac{7!}{5!}$.

   $\frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 7 \cdot 6 = 42$

   (c) In how many ways can 7 songs be put in order on a playlist?

   $7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5{,}040$ ways

10. **I count permutations with ${}_nP_r$, and I tell when repeats are allowed.**

   (a) 9 students run for president, vice president, and secretary. In how many ways can the 3 jobs be filled?

   Order matters. ${}_9P_3 = \frac{9!}{6!} = 9 \cdot 8 \cdot 7 = 504$ ways

   A locker code uses 4 letters from A to F (6 letters).

   (b) How many codes are there if letters can repeat?

   $6 \cdot 6 \cdot 6 \cdot 6 = 6^4 = 1{,}296$ codes

   (c) How many codes are there if no letter repeats?

   ${}_6P_4 = 6 \cdot 5 \cdot 4 \cdot 3 = 360$ codes

6.6 - Combinations

11. **I count combinations with ${}_nC_r$.**

(a) Find ${}_6C_4$.

$$\frac{6!}{4!\,2!} = \frac{6 \cdot 5}{2 \cdot 1} = 15$$

(b) 11 teams are in a league. Each team plays every other team once. How many games are there?

A game is a pair of teams. ${}_{11}C_2 = \frac{11 \cdot 10}{2} = 55$ games

(c) You choose 4 of 9 photos for a poster. In how many ways can you choose them?

${}_9C_4 = \frac{9 \cdot 8 \cdot 7 \cdot 6}{4 \cdot 3 \cdot 2 \cdot 1} = 126$ ways

12. **I decide whether a problem needs a permutation or a combination, and I explain the difference.**

For each situation, write permutation or combination. Then find the number of ways.

Situation	**Permutation or combination?**	**Number of ways**
(a) Choose 3 of 11 students for a clean-up crew.	combination	${}_{11}C_3 = 165$
(b) Give 1st, 2nd, and 3rd place to 3 of 11 science projects.	permutation	${}_{11}P_3 = 990$

(c) Why is the answer to (b) 6 times the answer to (a)?

Each group of 3 can be put in order 3! = 6 ways. A permutation counts every order. A combination counts the group once.

(d) Which situation is a combination? Circle the letter.

A Choosing a captain and a co-captain from 15 players.
(B) Choosing 5 of 15 players to ride in the first van.
C Arranging 5 trophies in a row on a shelf.
D Making a 4-digit code with no repeated digits.

6.7 - Probability with Counting

13. **I count the possible outcomes and the outcomes I want, and I find the probability.**

A coach picks 3 players at random from 7 forwards and 5 defenders.

(a) How many groups of 3 are possible?

$${}_{12}C_3 = \frac{12 \cdot 11 \cdot 10}{3 \cdot 2 \cdot 1} = 220 \text{ groups}$$

(b) Find P(all 3 are forwards).

${}_7C_3 = 35$ groups of forwards. $\quad \frac{35}{220} = \frac{7}{44} \approx 0.16$

(c) Find P(all 3 are defenders).

${}_5C_3 = 10$ groups of defenders. $\quad \frac{10}{220} = \frac{1}{22} \approx 0.05$

14. **I decide if order matters in a probability problem.**

15 students are in a club. Does order matter? Find the probability.

(a) 3 students are chosen at random to be president, vice president, and secretary. What is the probability that Amira is president, Ben is vice president, and Carla is secretary?

Yes, the jobs are different. ${}_{15}P_3 = 2{,}730$. $\quad P = \frac{1}{2{,}730} \approx 0.0004$

(b) 3 students are chosen at random for a committee. What is the probability that the committee is Amira, Ben, and Carla?

No, it is the same job. ${}_{15}C_3 = 455$. $\quad P = \frac{1}{455} \approx 0.002$